

Kod Projek :

BITU 3973

FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY

 PROJEK SARJANA MUDA 1 (BITU3973)
 PROPOSAL

A	TITLE OF PROPOSED PROJECT: UTeM Waste Disposal Management System
B	DETAILS OF STUDENT
	Name: MUHAMMAD FARIZ BIN ROHIZAD Matric No: B032210332 Program Pengajian: BITD Handphone No.: 011-6146 6216 E-mail Address: fxiryz07@gmail.com Semester/Session: Sem.2, 2024/2025
C	PROJECT INFORMATION
C(i)	Executive Summary of Project Proposal (maximum 300 words) As environmental concerns continue to grow, universities, including UTeM, must take proactive steps to address sustainability challenges. This project aims to develop an API-based Green Activity System that efficiently manages and organizes waste data at UTeM, supporting the university's goal of becoming an eco-friendly institution. Currently, UTeM lacks an effective waste data management system, making it difficult to track, analyse, and reduce waste. Manual processes involving Excel spreadsheets and inconsistent data entry result in errors and inefficiencies, hindering the effectiveness of recycling and waste reduction efforts. The project addresses these issues by developing a centralized system that enables CENSEI staff to input, track, and analyse waste data more effectively. The system automates waste data collection, reduces human error, and improves data accuracy, thus enabling better waste management practices and more accurate reporting. The system will categorize waste into types such as organic and non-organic, making it easier to measure recycling progress, track waste reduction, and reduce greenhouse gas emissions. Key features include a user management module for secure login and audit functionality, a waste disposal management module to view and update waste data, and a system management module to generate visual reports like graphs and pie charts. The system will also facilitate data backup and analysis, allowing UTeM to monitor its sustainability efforts and identify areas for improvement.

C(ii) Detailed proposal of project:**(a) Project background including Introduction / Problem Statements.**

As environmental challenges become increasingly urgent, universities must take proactive steps. This project aims to develop a Green Activity System for UTeM, leveraging an API-based approach to streamline and enhance eco-friendly initiatives. The primary objective is to create an effective system for managing and organizing waste data at UTeM. The system will categorize waste into types such as organic and non-organic, making it easier to monitor recycling efforts and reduce greenhouse gas (GHG) emissions on campus. Using the API, staff will be able to collect, analyse, and share waste data from various locations across the university, supporting UTeM's transition to a more sustainable environment.

This project is crucial in advancing UTeM's goal of becoming a greener institution. The system will generate clear, accurate waste management reports, allowing the university to track its progress. It will also enable more efficient waste categorization and recycling program management, helping to reduce overall waste. Furthermore, the system will identify areas for improvement, making waste management more effective.

In addition, this system could serve as a model for other universities seeking to enhance their sustainability efforts through data-driven solutions. By setting an example, UTeM can inspire other institutions to adopt intelligent and sustainable waste management practices, contributing to a cleaner and healthier environment for all.

Problem statements

1. **Lack of a system for waste data management:** Right now, CENSEI doesn't have a system to manage waste disposal data, which makes it difficult for staff to enter, track, and analyse waste information efficiently.
2. **No centralized platform for data:** Since there is no single system, it's hard to visualize waste data, such as recycling progress or waste reduction, using charts or reports. This makes it challenging to measure how well recycling is working or to track how much waste is being reduced, which makes it harder to lower greenhouse gas emissions effectively.
3. **Manual process for waste disposal:** Faculty members must manually enter the detail of waste and estimated weight of waste into an Excel spreadsheet after taking images of it. The data must be reorganized and rectified if errors are discovered once this file is verified by CENSEI personnel. Additionally, CENSEI employees must enter the information that has been sent to them in their Excel spreadsheet in the format that their management has requested. This manual procedure takes a lot of time and frequently results in errors, which reduces the effectiveness of waste management.

(b) Objective (s) of the Project

This project embarks on the following objectives:

1. To develop a web-based system to allow CENSEI staff to easily input, track, and analyze waste data for more efficient waste management

2. To create a centralized platform using web-based to track recycling efforts, measure waste reduction progress, and reduce greenhouse gas (GHG) emissions more effectively and easily.
3. To automate the waste data collection and verification process by integrating an API-based system that reduces human error, saves time, and improves data accuracy in waste management.

(c) Scope of the Project

Users -

CENSEI staff

- Have privilege to manage data of waste and do waste management operations such view, insert, delete, search and update on this system.
- Have privilege to view the report on waste disposal data in the tabulated form such as graph and pie chart.
- Auditing the user that log in this system.

Modules -

- **User Management Module**

This module is for CENSEI staff to need to register first before logged in the system. CENSEI staff need to key in their information detail such as name, age, phone number. Moreover this module is for CENSEI staff to auditing the user that logged in this system such as get what the of user, what operation of the data that user doing and what time the user log in.

- **Inorganic and Organic Waste Management Module**

This module deals with the management of waste disposal. It contains all the inorganic and organic waste data, which will be shown according to its organic and non-organic categories. Inorganic waste data contain of paper, plastic, metal, other class, fabric and e-waste. E-waste has two type of data which is toxic and non-toxic. Next organic data contain of waste cooking oil and compost. The detail of organic and inorganic consist of waste number, type of waste, location the waste taken and weight. All of data about inorganic and organic waste will be fetch directly on the database and will be show in table on the webserver. Moreover, CENSEI staff can view the total inorganic and organic waste in pie chart and the graph of weight of organic and inorganic by month in selected year. In this module, CENSEI also can change the weight of inorganic and organic waste if the estimated weight differs from the actual weight after the recycling centre staff waste's weight, has measured the weight. Additionally, CENSEI employees have the ability to remove waste disposal data that is deemed rejected. Next CENSEI staff can also search of inorganic and organic waste based on it type and location of the waste taken.

- **Hazardous Waste Module**

This module deals with the management of waste disposal. It contains all hazardous waste data. Hazardous waste data contain of bulb and lamps, lubricant, hydraulic, transformer oil (natural + mineral), sanitizing, disinfection, discarded drug, empty disposed container, contaminated rag or glove, obsolete and laboratory chemical. All this that data will be taken from different places. The detail of hazardous waste consists of waste number, type of waste, location the waste taken, where it stores and its weight. All of data about hazardous waste will be fetch directly on the database and will be show in table on the webserver. Moreover, CENSEI staff can view the hazardous waste in graph total weight by location. In this module,

	CENSEI also can change the weight of hazardous waste if the estimated weight differs from the actual weight after the recycling centre staff waste's weight, has measured the weight. Additionally, CENSEI employees have the ability to remove waste disposal data that is deemed rejected. Next CENSEI staff can also search of hazardous waste based on it type and location of the waste taken. Hardware – • Mouse • Laptop Software – • Postgress • Xampp • Php • Visual Studio Code • Node.Js • Postman (d) Expected Outcome/ Proposed Solution The project's anticipated result is the effective deployment of a centralized, API-based Green Activity System at UTeM, which will greatly increase data accuracy and expedite waste management procedures. More successful recycling and trash reduction initiatives will result from this system's ability to track, classify, and report garbage in an efficient manner. UTeM will be able to better monitor its sustainability efforts thanks to the system's accurate, real-time insights into waste management methods, which will be provided by automating data gathering and minimizing human error. Making visual reports, such pie charts and graphs, will help people make well-informed decisions and point out areas that need work.
D	REFERENCES
	State your references (Minimum 5 references- report, books, URLs) 1. Universiti Malaya Sustainable Development Centre. (2023). UM waste management report 2022. Universiti Malaya. https://sustainability.um.edu.my/doc/Sustainability%20UM/Sustainability%20Reporting/UM%20Waste%20Management%20Report%202022.pdf 2. Rogers, H. (2005). <i>Gone tomorrow: The hidden life of garbage</i>. 3. Pusat Alam Sekitar Pintar. (n.d.). CENSEI UTeM. Retrieved February 23, 2025, from https://censei.utem.edu.my/index.php/ms/ 4. (2018 onwards. Source: DEFRA, 2018) 5. CleanHub. (2024, November 5). <i>What is waste management?</i> CleanHub Blog. https://blog.cleanhub.com/what-is-waste-management

E (i)	Declaration by student / Akuan pelajar
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	Date : 16 MARCH 2025 Tarikh : _____	Student's Signature : Tandatangan Pelajar : 
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E (ii)	Recommendation by the Supervisor <i>Perakuan oleh Penyelia</i>	Recommendation by the Committee <i>Perakuan oleh Jawatankuasa</i>
	Select: Recommended / Not Recommended Comments: Good proposal with clear scope. Supervisor's Name: ROSLEEN BINTI ABDUL SAMAD Signature and Stamp:   Date: 16/3/2025 Tarikh:	Select : Accepted / Not Accepted Comments: Committee's Name: Signature and Stamp: Date: Tarikh: